

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)**  
**ACADEX PRACTICE EXAMINATION**  
**A-LEVEL · FURTHER MATHEMATICS**

**FURTHER MATHEMATICS 9187/2**

**PAPER 2 November 2024**

**3 hours**

**PAPER 2 — STRUCTURED QUESTIONS — CALCULATOR ALLOWED**

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers. This script is unique to November 2024 9187/2.

**Additional materials:** A calculator may be used unless a question forbids it.

**INSTRUCTIONS:** Answer all questions. Full marks are not given for answers only — show method.

1. A particle is projected at 20 m/s at  $30^\circ$  to the horizontal. Take  $g = 10 \text{ m/s}^2$ . Find time of flight on level ground and range. **[10]**
2. A 4 kg mass on a smooth plane inclined at  $30^\circ$ . Find acceleration down the plane ( $g=10$ ). **[10]**
3. A uniform beam 4 m, mass 6 kg, supported at both ends. A 9 kg mass is 1 m from A. Find reactions ( $g=10$ ). **[10]**
4.  $X \sim B(5, 1/3)$ . Find  $P(X=2)$ . **[8]**
5.  $X \sim N(50, 16)$ . Find  $P(X > 54)$  in terms of  $\Phi$ , using  $Z=(X-\mu)/\sigma$ . **[8]**
6. Calls arrive at 3 per hour. Using Poisson, write  $P(\text{exactly 2 in an hour})$ . **[8]**
7. Further Statistics: Hypothesis test outline. Give definitions and one calculation. **[8]**
8. Further Statistics: Correlation interpretation. Give definitions and one calculation. **[8]**
9. Further Statistics: Discrete random variable  $E(X)$ . Give definitions and one calculation. **[8]**
10. Further Statistics: Continuous pdf  $f(x)=k$  on  $[0,2]$ . Give definitions and one calculation. **[8]**

**11.** Further Statistics: Sampling / CLT one-liner. Give definitions and one calculation. **[8]**

**12.** Further Statistics: Chi-square idea. Give definitions and one calculation. **[8]**

**END OF QUESTION PAPER**

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).